

軟體測試與資安檢測

教師：王凡

授課時間：每週三 9：10—12：00

博理館 113

課程代號：EE 5170

識別碼：921 U9550

民國 110 年春季班

QUIZ (2021/03/31)

Open-book, open-internet, no discussion

Name: _____ Department : _____ Student ID: _____

1. Suppose we have the following methods. Please list all the interprocedural DU-pairs.

```
def main (x,y):  
    z = x + y  
    a = scroll(z, y)  
    a = x + a  
    b = scroll(a, x)  
    return a + b  
  
def scroll (u,v):  
    print (u)  
    x = u + v  
    if u < 100:  
        return x  
    else:  
        return v
```

2. Assume that we have a class person with three methods, addName(), addAddress(), and register() subject to the following sequence constraints:

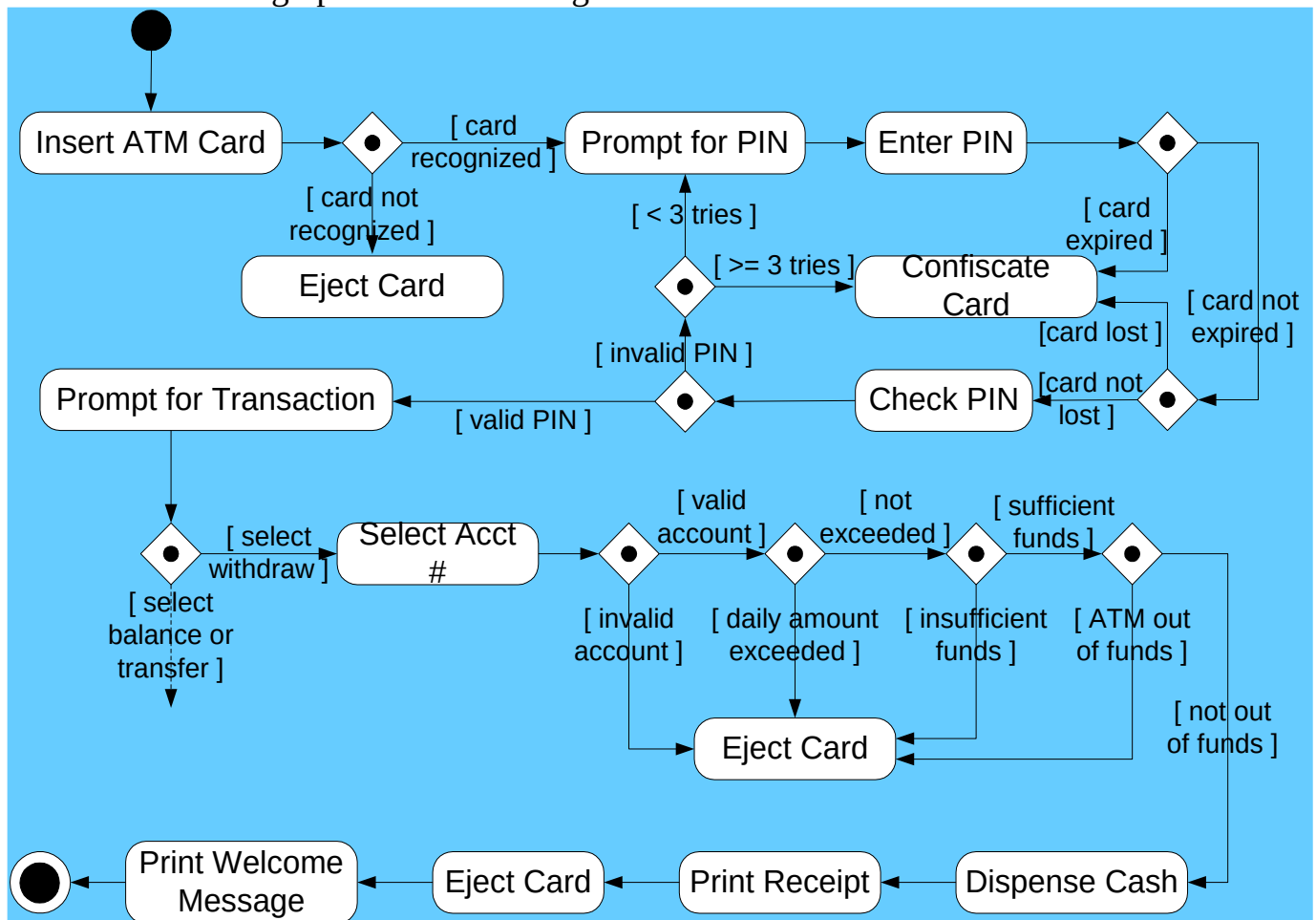
- a) A person object must be instantiated to use any of the three methods.
- b) Only after a person object has executed addName() and addAddress() each at least once can the object execute the register() method.

Please draw a control flow graph that calls this class and test this class for correct implementation of the sequence constraints.

3. Continuing from problem 2, please show a set of test paths that test the modules for correct usage of the class with full node coverage. You must explain why your answer is correct.

4. Continuing from problem 2, please show a set of test paths that test the modules for incorrect usage of the class with full node coverage. You must explain why your answer is correct.

5. Consider the state graph in the following.



Please design two scenarios to test the system in two exception handling use cases.